

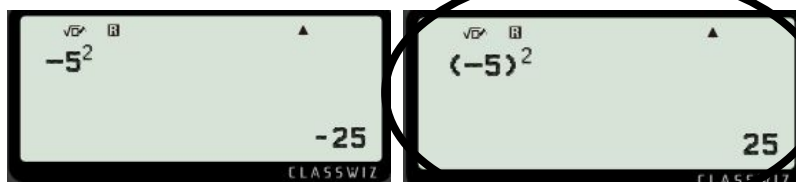


Limitations of technology in calculations

Task A: Correct inputs

1) Circle the correct and/or accurate calculation on the calculator display for the following:

a) The square of negative 5



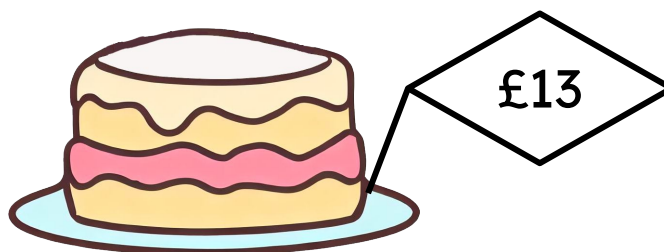
b) $0.3333... + 0.3333... + 0.3333...$



c) 5 multiplied by $\frac{3}{4}$



2) Andeep, Jun and Jacob want to buy a cake and divide the amount equally between them.



Each student pays the cake shop manager £4.33 each.

Explain why the cake shop manager is not going to give them the cake.

*3 lots of £4.33 does not sum to £13.
They have not paid the full price. They are 1p short.*



Task B: Calculator display and limitations

1) Identify if the statement is true or false when using the scientific calculator Casio Classwiz fx 991.

Statement	True	False
The calculator will convert 0.2673918 into a fraction automatically.		✓
It is more accurate to use 0.16666 rather than $\frac{1}{6}$		✓
Inputting a recurring digit to 18 decimal places will convert it to a fraction.	✓	
The priority of operations is applied to all calculations.	✓	

2) Andeep has discovered a way to find the fractional form to a decimal when the calculator does not do it automatically.



- a) What is the fractional form of 0.346774?
- b) What is the simplified fractional form of 0.346774?

3) A calculator displays this number 0.0539562635

- a) Give three examples, in decimal form, of what this number could be if it were a recurring decimal.
- b) Give an example of a fraction it could be if it were a terminating decimal.

a) Some examples of a non-terminating decimal

0.0539562635454545454...
 0.05395626348888888...
 0.053956263511111111..

b) It could be this terminating decimal

$$\frac{539562635}{10\,000\,000\,000}$$